

Frederick house sales 2019

Daniel Coblentz

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Packages

In addition to our usual `tidyverse` package, we load

- `scales` for making nice axis labels with variables measured in dollars. Use + `scale_x_continuous(labels = label_dollar())`.
- `broom` for using models to “augment” data sets with fitted values and residuals. Check your notes/posted class notes for details.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(broom)
```

The data

We will study a dataset that includes records of sales of residential properties in Frederick County from 2019.

The next chunk loads the data and converts two of the variables from text to categories.

```
frederick_2019 <- read_csv("data/frederick_2019.csv") %>%
  mutate(basement = factor(basement, levels = c("NO", "YES", "FINISH")),
         heat = fct_infreq(heat))

## Rows: 3284 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (2): basement, heat
## dbl (6): price, land_area, building_area, full_baths, half_baths, condition
## lgl (1): has_fireplace
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Brief codebook

The cases in the data set are the sales of residential properties in Frederick County in 2019. If a property was sold multiple times in 2019, only the last sale is recorded. A relatively small number of sales (on the order of 100) were removed from the dataset since their records were incomplete.

Variables

Read the definitions of the variables carefully.

- **price**: the sales price of the property
- **land_area**: the area of the land associated with the property, measured in acres
- **building_area**: the area of the main building on the property, not including “below grade” space such as a basement.
- **heat**: the type of heat for the main building
- **full_bath**: the number of full baths (bathrooms with bathing facilities) in the main building
- **half_bath**: the number of half baths (bathrooms without bathing facilities) in the main building
- **condition**: condition of the main building on an 8-point scale, where 8 is “excellent”
- **has_fireplace**: whether the main building has at least one fireplace

###0. Examine the dataset in the usual ways and familiarize yourself with the variables and their types.

```
head(frederick_2019, 5)
```

```
## # A tibble: 5 x 9
##   price land_area building_area basement heat full_baths half_baths condition
##   <dbl>   <dbl>       <dbl> <fct>   <fct>   <dbl>   <dbl>   <dbl>
## 1  50000    0.74         1600 YES     SPACE     1       0       3
## 2  75000    0.25          992 YES     SPACE     0       0       2
## 3 300000    0.43         1080 FINISH  ELEC     2       1       4
## 4  40000    0.12          880 NO      SPACE     1       0       2
## 5 332000    0.22         2080 YES     HT PMP    2       1       4
## # i 1 more variable: has_fireplace <lgl>
```

```
glimpse(frederick_2019)
```

```
## Rows: 3,284
## Columns: 9
## $ price      <dbl> 50000, 75000, 300000, 40000, 332000, 318500, 265000, 425~
## $ land_area  <dbl> 0.7400, 0.2500, 0.4300, 0.1200, 0.2200, 2.4200, 1.3900, ~
## $ building_area <dbl> 1600, 992, 1080, 880, 2080, 1298, 1092, 2136, 1196, 1804~
## $ basement   <fct> YES, YES, FINISH, NO, YES, YES, YES, YES, YES, YES, ~
## $ heat       <fct> SPACE, SPACE, ELEC, SPACE, HT PMP, HT PMP, HT PMP, HT PM~
```

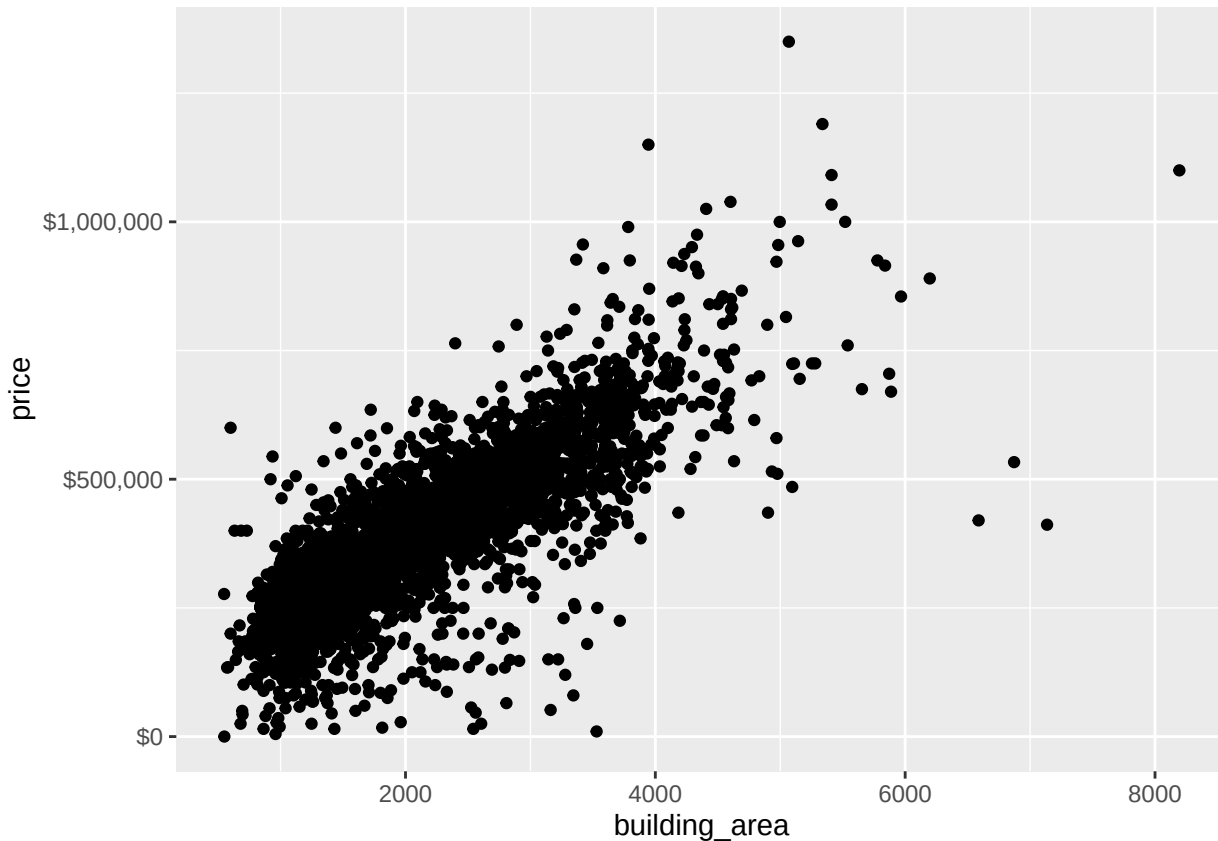
```
## $ full_baths    <dbl> 1, 0, 2, 1, 2, 2, 2, 3, 1, 2, 2, 1, 1, 2, 1, 1, 2, 3, 1, ~
## $ half_baths    <dbl> 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 1, ~
## $ condition     <dbl> 3, 2, 4, 2, 4, 4, 4, 4, 4, 3, 4, 4, 3, 4, 3, 4, 4, 4, 5, 5, ~
## $ has_fireplace <lgl> TRUE, FALSE, TRUE, FALSE, FALSE, TRUE, TRUE, TRUE, TRUE, FALSE~
```

A model for predicting sales price

Today, we will use `building_area` to predict the price of a house.

1. Create a basic scatterplot of price vs `building_area`.

```
ggplot(data = frederick_2019,
       mapping = aes(x = building_area,
                     y = price)) +
  geom_point() +
  scale_y_continuous(labels = label_dollar())
```

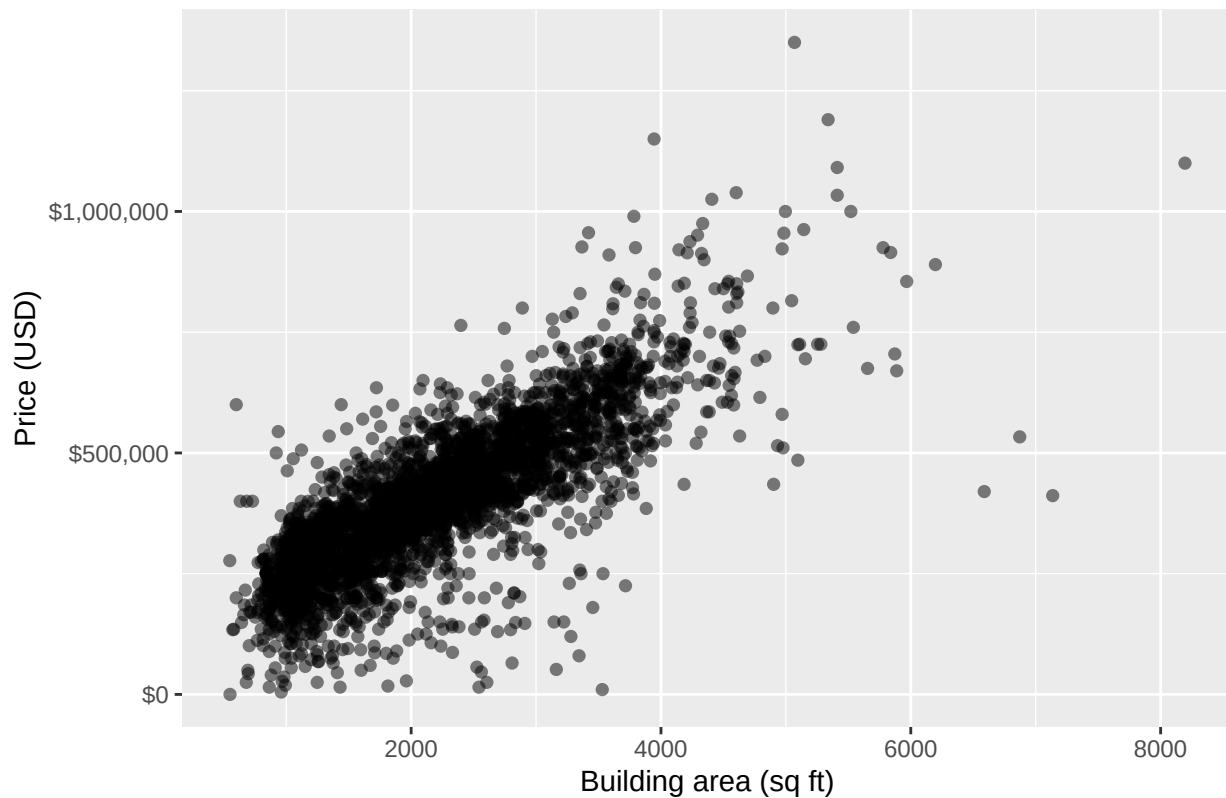


2. Improve your plot: Address overplotting in the usual ways using `shape`, `size`, and `alpha`. Add axis labels and a title. Also, add an extra layer to the plot to make the y-scale print in a more user-friendly way. The extra layer uses the `scales` package and is: `+ scale_y_continuous(labels = label_dollar())`

```
ggplot(data = frederick_2019, mapping = aes(x = building_area, y = price)) +
  geom_point(shape = 16, size = 2, alpha = 0.5) +
  scale_y_continuous(labels = label_dollar()) +
  labs(title = "Price vs building area with overplotting adjustments",
```

```
x = "Building area (sq ft)",
y = "Price (USD)"
```

Price vs building area with overplotting adjustments



Hopefully, you agree that it looks reasonable to model this relationship using a linear model.

2. Use `lm` to create a linear model for this relationship. Make sure you have the variables the right way around!

```
fit <- lm(price ~ building_area, data = frederick_2019)
summary(fit)
```

```
##
## Call:
## lm(formula = price ~ building_area, data = frederick_2019)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -659019  -46105    4204    53819   559236
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.045e+05  4.240e+03   24.64  <2e-16 ***
## building_area 1.354e+02  1.759e+00   76.98  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 94690 on 3282 degrees of freedom
```

```
## Multiple R-squared:  0.6436, Adjusted R-squared:  0.6435
## F-statistic:  5926 on 1 and 3282 DF,  p-value: < 2.2e-16
```

3. Print the model summary output.

above

4. Use the model summary output to write the equation of the model line. You will probably want to convert the slope and intercept, which are currently written in scientific notation, to decimal notation.

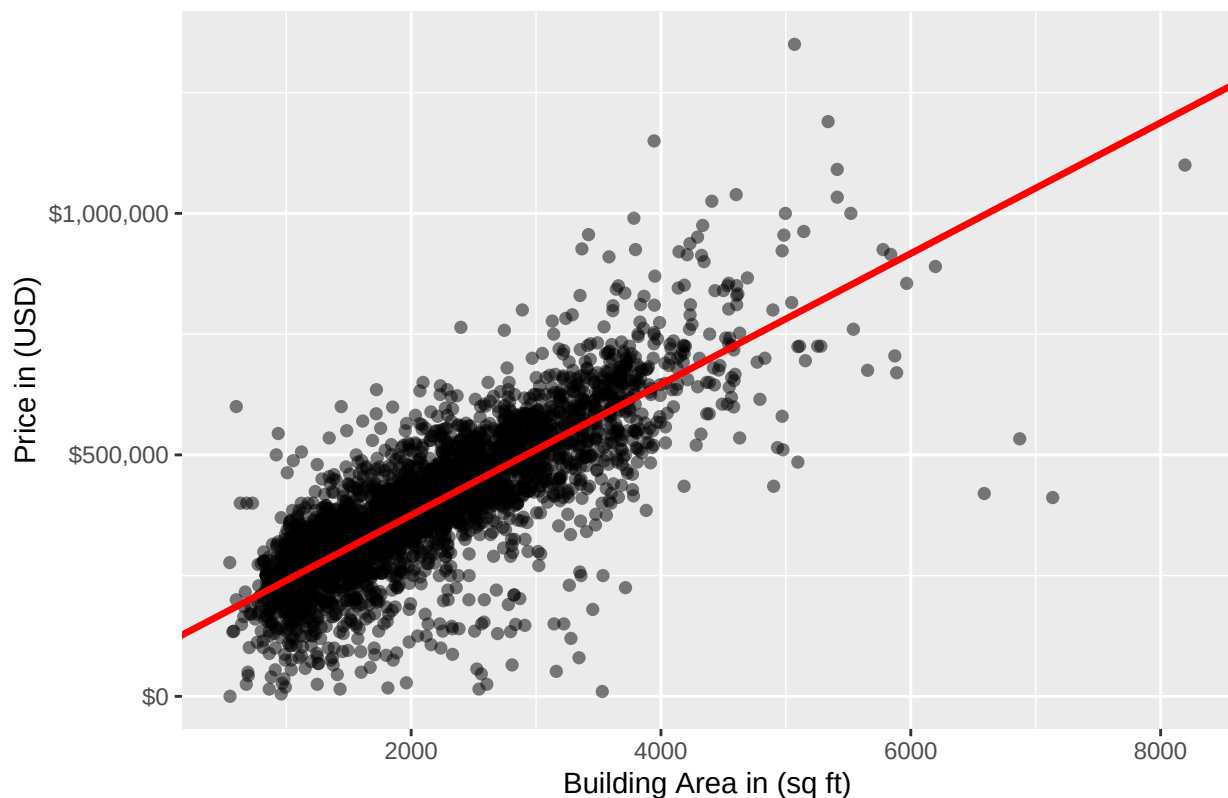
$\text{Price}_{\text{hat}} = 104500 + 135.4 * \text{BuildingArea}$

4a. Graph the original data with the model.

```
ggplot(data = frederick_2019, mapping = aes(x = building_area, y = price)) + geom_point(shape = 16, size = 1) +
  geom_abline(intercept = 104500, slope = 135.4, color = "red", size = 1.2) + scale_y_continuous(labels =
    labs(title = "Price vs building area with model equation",
      x = "Building Area in (sq ft)", y = "Price in (USD)"))
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

Price vs building area with model equation



5. Write an interpretation for the intercept of your model equation, in context, as we have been practicing.

Intercept: If a house had zero square feet, the predicted price would be \$104,500. However this does not make sense because no house has zero square feet although it does represent the theoretical base price before adjusting for building area.

6. Write a sentence of interpretation for the slope of your model line, in context, in the way that we have been practicing.

The slope is \$135.4 meaning for each additional square foot of building area, the predicted sales price increases by \$135.40.

Next, we will assess the model. We have already agreed that it looks like the linearity condition holds.

7. Augment the data frame with the model summary output. You should be using `augment()` here.

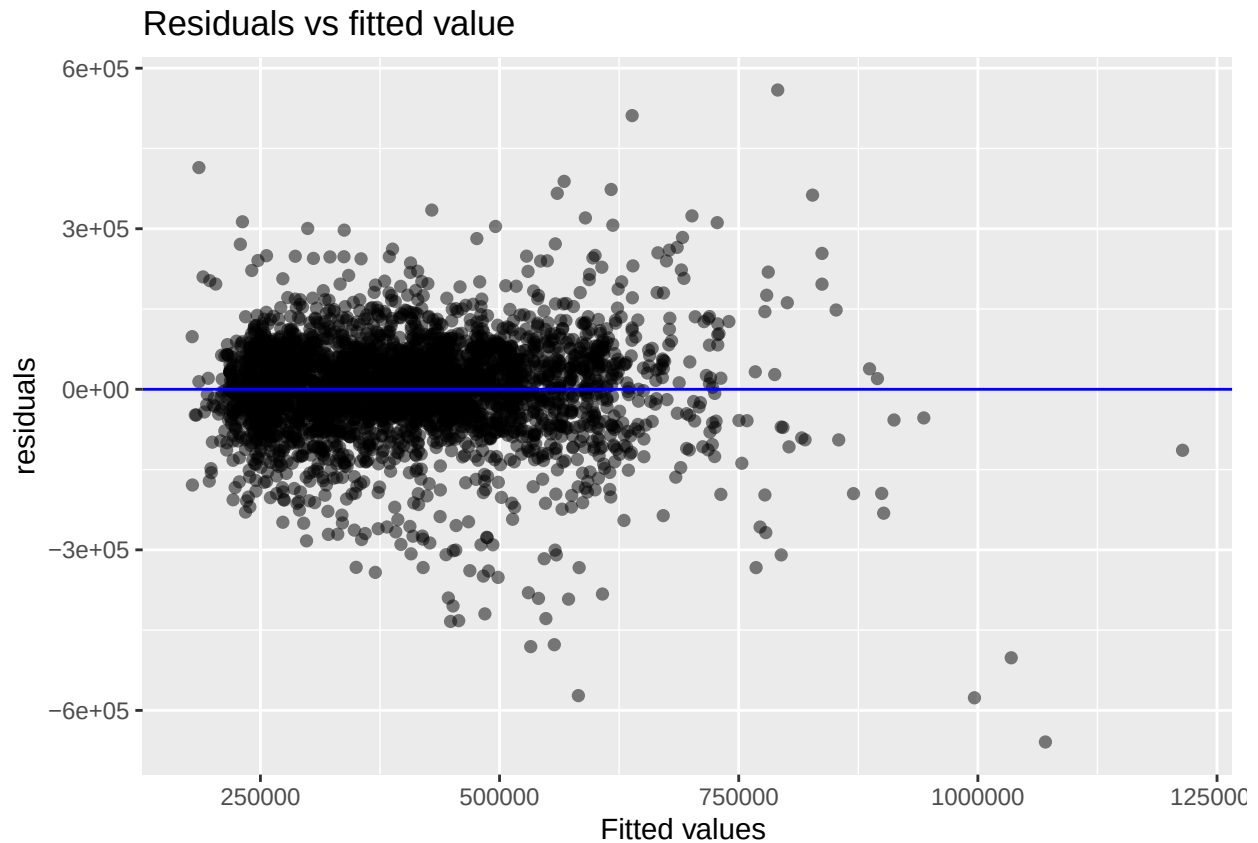
```
augmented_data <- augment(fit)
head(augmented_data)
```



```
## # A tibble: 6 x 8
##   price building_area .fitted .resid .hat .sigma .cooksd .std.resid
##   <dbl>         <dbl>   <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
## 1  50000          1600 321089. -271089. 0.000437 94586. 0.00179   -2.86
## 2  75000           992 238771. -163771. 0.000825 94661. 0.00124   -1.73
## 3 300000          1080 250685.  49315. 0.000753 94700. 0.000102    0.521
## 4  40000           880 223607. -183607. 0.000924 94650. 0.00174   -1.94
## 5 332000          2080 386077. -54077. 0.000311 94699. 0.0000508  -0.571
## 6 318500          1298 280201.  38299. 0.000598 94702. 0.0000490    0.405
```

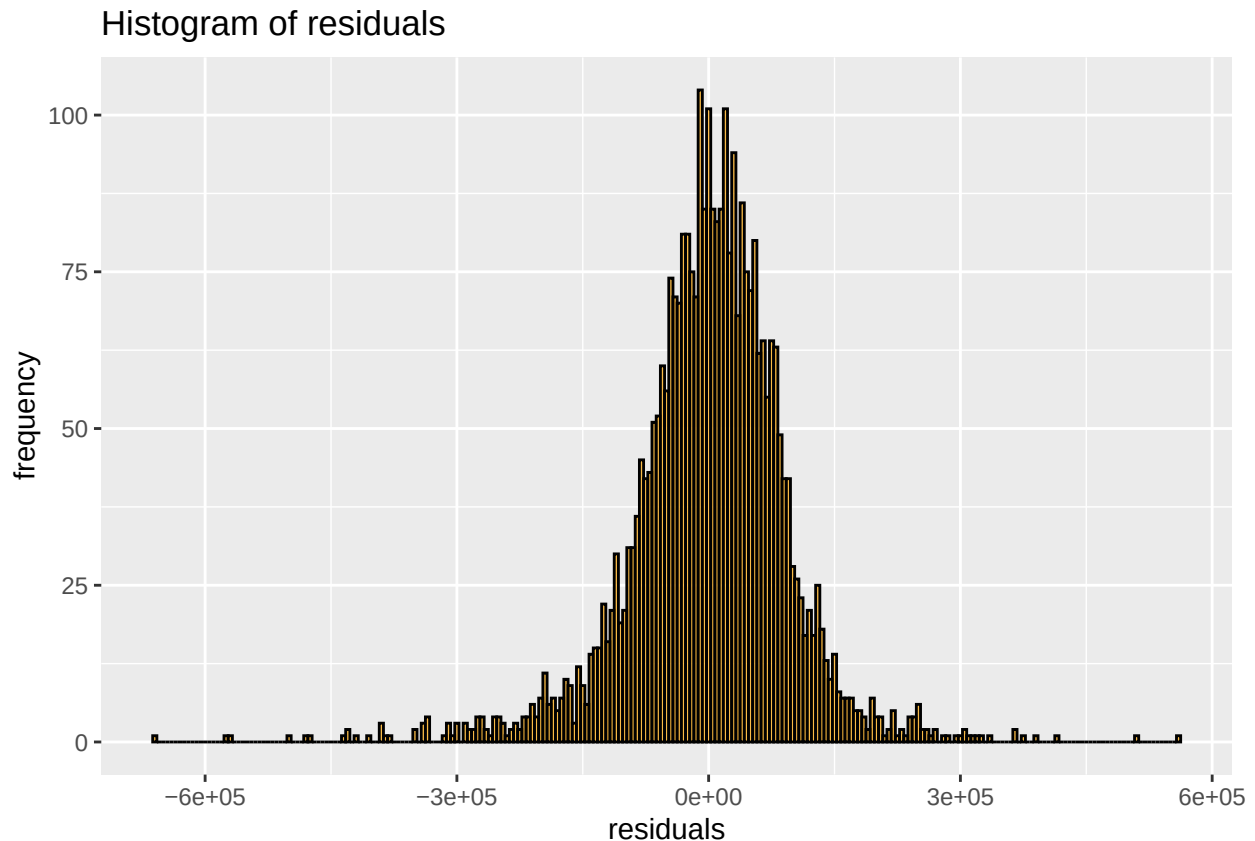
8. Next, make a scatterplot of the residuals versus the fitted values. Include a horizontal line at $y=0$. Does the graph look at the way it is supposed to look? Explain.

```
ggplot(data = augmented_data, aes(x = .fitted, y = .resid)) +
  geom_point(shape = 16, size = 2, alpha = 0.5) +
  geom_hline(yintercept = 0, linetype = "solid", color = "blue") +
  labs(title = "Residuals vs fitted value", x = "Fitted values ", y = "residuals")
```



Yes, the graph generally looks how we would expect. As most of the residuals are centered around the horizontal line at $y = 0$. Additionally there are some very large residuals (both positive and negative) far from zero indicating some outliers in the dataset. While the linear model seems reasonable overall some data cleaning to possibly remove outliers might improve our result. ### 9. Finally, make a histogram of the residuals. Does it look the way it is supposed to look? Explain.

```
ggplot(data = augmented_data, aes(x = .resid)) +  
  geom_histogram(binwidth = 5000, color = "black", fill = "orange", alpha = 0.7) + labs(title = "Histogram of residuals")
```



The histogram of residues seems correct as it has a bell shape with a peak around zero meaning most errors are evenly spread out across the quantiles of the distribution.

10. What does the model predict for the sales price of a house in Frederick that has 2070 square feet of building area?

```
predicted_price <- 104500 + (135.4 * 2070)
predicted_price
```

```
## [1] 384778
```

The model predicts a sales price of \$384778 for a home in frederick that has 2070 square feet of building area.

11. There is a house in the data set that has 2070 square feet of building area and a sales price of \$285,000. Calculate the residual for this house. What does the sign of the residual tell you about the data point?

```
actual_price <- 285000
residual_value <- actual_price - predicted_price
residual_value
```

```
## [1] -99778
```

The residual is negative which means the actual sale price was lower than the predicted price and suggests a model overestimated the price for this home.

12. (AFTER WE DISCUSS R and R²) Find the R² value in the model output, and write a sentence that explains what that R² value tells you about the relationship between these two variables.

The r² value of 0.6436 indicates a 64.36% of the variability in house prices can be explained by building area. This suggests that building area is a strong predictor of house prices however other factors also contribute.

###13. Take the square root of that (and adjust the sign if necessary) to find the r value for these two variables. What does that tell you about the strength of the correlation between them?

$r = \sqrt{r^2}$ $r = \sqrt{0.6436} = 0.802$ 0.8022 Since our slope in the scatter plot is positive R will also be positive.

The correlation coefficient r calculated above suggests a strong positive Correlation between building area and housing price. This means that as building price increases housing prices tend to increase as well.